Rethinking Policy Optimization with Gradient Estimators

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Abstract

Clipping is widely used in policy optimization to stabilize learning under policy mismatch. In large language model post-training, however, importance-weighted updates with weak or no clipping can learn effectively for many iterations before abruptly becoming unstable, a behavior not explained by population-level policy-deviation arguments alone. We show that this transition can be understood as a finite-sample gradient-estimation problem. By expressing the off-policy policy gradient as an importance-weighted mean estimator, we connect its estimation error to effective sample size (ESS) and establish how this error determines finite-sample policy improvement. This analysis further reveals when the variance reduction from truncation or clipping outweighs the useful gradient signal it removes, motivating a simple ESS-conditioned rule that activates clipping only after estimator reliability deteriorates. Controlled experiments and large-scale LLM training show that ESS predicts this reliability transition and that ESS-conditioned clipping improves stability and learning efficiency across model scales and objectives.

1 Introduction

Reinforcement learning has become central to the post-training of large language models (LLMs) [1, 13, 15]. In scalable training systems, policy optimization is often effectively off-policy: a rollout policy generates responses, while a trainer reuses the collected data for one or more gradient updates. Consequently, the current policy need not coincide with the policy that generated the data [9]. This mismatch makes it important to control how far policy optimization moves from the rollout policy.

Controlling such policy deviation has long been a central principle in policy optimization. Trust Region Policy Optimization (TRPO) formalizes this principle through population-level policy improvement guarantees, while Proximal Policy Optimization (PPO) provides a simple and scalable approximation through ratio-based clipping [11, 12]. PPO-style objectives, including GRPO, inherit this intuition by clipping or masking updates when sampled likelihood ratios move too far from one [13]. The resulting theoretical picture is clear at the population level: if population gradient estimators are available, keeping the updated policy sufficiently close to the rollout policy controls the error of the surrogate objective and thereby protects policy improvement.

LLM training, however, presents a striking tension with this population-level view. Empirically, permissive estimators, including truncated importance sampling (TIS) with large caps or even unclipped updates, can remain stable and learn considerably faster. Yet this success need not persist. Figure 1 illustrates this phenomenon: TIS with caps 3 and 5 improves steadily for many updates before abruptly collapsing. The same update rule can therefore be highly effective for a long period and suddenly become unreliable. This raises the central question of this paper: *when does a permissive policy-gradient estimator cease to provide a reliable update?*

Classical trust-region theory explains why moving too far from the rollout policy can be unsafe [9, 11, 12]. It does not, however, explain the delayed transition in Figure 1, where failure follows

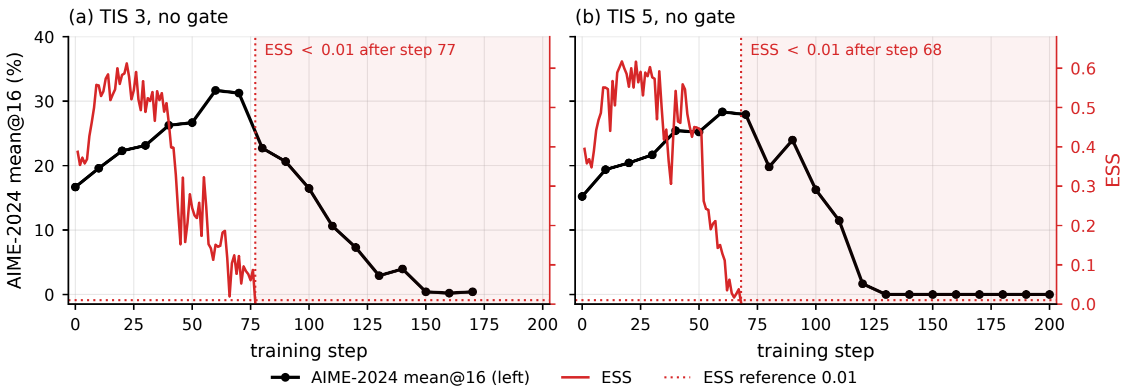

Figure 1: Delayed failure under permissive TIS updates on Qwen3-30B-A3B. TIS with caps 3 and 5 improves AIME-2024 mean@16 for many updates before effective sample size (ESS) deteriorates and performance subsequently collapses. The delayed transition motivates our central question: when does a finite rollout cease to support a reliable policy-gradient update?

many apparently successful updates rather than a single conspicuous policy shift. Repeated updates can progressively concentrate complete-trajectory importance weights, reducing the effective support provided by a finite rollout even when the empirical objective is constructed from factorwise coefficients. A population-level bound controls policy deviation, but it does not characterize whether the available rollout still contains enough effective information to estimate the next update reliably. The relevant question is therefore not only *how far the policy has moved*, but also *whether the finite rollout still supports a reliable gradient estimate.*

In this paper, we resolve this tension by shifting the analysis from population-level policy deviation to finite-sample update reliability. We show that the reliability of the policy-gradient estimator plays a central role in determining whether an empirical update translates into policy improvement. This perspective recasts clipping as a form of estimator control: permissive estimators can preserve more of the learning signal when the rollout provides sufficient effective support, whereas controlling the estimator becomes increasingly valuable as that support deteriorates. We make this connection explicit through effective sample size (ESS), which quantifies the concentration of the importance weights underlying the finite-sample estimator. We then validate this prediction empirically and use it to derive a simple optimization rule that preserves permissive updates while they remain reliable and intervenes when the rollout ceases to support them.

Our contributions are threefold. **Finite-Sample Policy Improvement:** We derive a policy-improvement analysis that explicitly accounts for the reliability of the empirical gradient estimator, complementing the population-level view of policy deviation with the statistical adequacy of the available rollout. **Estimator Reliability and ESS:** We interpret clipping and truncation as forms of estimator control and establish a direct connection between gradient-estimator reliability and effective sample size (ESS), explaining why permissive updates can learn rapidly when effective support is sufficient but become unstable as importance weights concentrate. **Reliability-Aware Optimization:** Guided by this analysis, we propose a simple optimization rule that preserves permissive updates while the rollout remains reliable and increases control only when reliability deteriorates. Controlled simulations and large-scale LLM reasoning experiments validate these predictions and demonstrate an improved stability-learning tradeoff over fixed clipping strategies.

2 Related work

Trust-region methods control policy deviation to obtain monotonic-improvement guarantees, while PPO replaces the explicit trust-region constraint with a ratio-clipped surrogate that is easier to optimize at scale [11, 12]. However, recent analyses of LLM policy optimization show that the trust region alone does not determine the reliability of policy optimization. For example, recent empirical analyses reveal that less aggressive clipping thresholds [5, 7, 9, 15] and even unclipped learning can be effective. Yet, existing works lack theoretical explanations.

Our analysis addresses a different source of error: the finite-sample gradient estimate used to make that update by linking the policy optimization to the importance-weighted estimation problem. Existing work in importance sampling reveals the central role of ratio moments in quantifying estimation uncertainty and applies this to the off-policy estimation problem [3, 4]. We further extend these two ingredients by relating importance-weight concentration to gradient-estimation reliability and, in turn, to policy improvement [6, 8].

3 Preliminaries

3.1 Policy gradients from a fixed rollout

Policy gradients. Let $X \sim \nu$ denote a context and let $P_\theta(\cdot \mid X)$ be a parameterized policy. For a response or trajectory $Y = (Y_1, \dots, Y_T)$ with reward $R(X, Y)$, the policy objective is given as $J(\theta) := \mathbb{E}_{X\sim\nu,\, Y\sim P_\theta(\cdot|X)}[R(X, Y)]$.

Parameterized policy learning seeks a parameter θ that maximizes the policy objective $J(\theta)$. Policy-gradient methods learn this parameter through gradient ascent. By classical control-variate analysis and the score function identity, the policy gradient admits the following form:

$$g(\theta) := \nabla J(\theta) = \mathbb{E}_{P_\theta}\left[A(X, Y)\sum_{t=1}^{T} \nabla_\theta \log P_\theta(Y_t \mid X, Y_{<t})\right]. \tag{1}$$

Here, $A(X, Y) := R(X, Y) - b(X)$ is the control-variate adjusted advantage, where the baseline $b(X)$ leaves the expected gradient unchanged while reducing its variance.

The expectation in Equation (1) is taken under the current policy P_θ. For efficiency, modern policy-optimization methods often rely on reusing data [11, 12], which creates a mismatch between the current policy and the rollout policy $Q(\cdot \mid X)$. This mismatch is then accounted for by the likelihood ratio.

Formally, define the likelihood ratio and score $r_{\theta,t} := \frac{P_\theta(Y_t|X,Y_{<t})}{Q(Y_t|X,Y_{<t})}$, $\ell_{\theta,t} := \nabla_\theta \log P_\theta(Y_t \mid X, Y_{<t})$. Given i.i.d. rollouts $Z_i = (X_i, Y_i) \sim \nu(dx)Q(dy \mid x)$, $i = 1, \dots, N$, and assuming $P_\theta(\cdot \mid X) \ll Q(\cdot \mid X)$, a change of measure rewrites the population gradient and its sample analogue under the rollout distribution:

$$g(\theta) = \mathbb{E}_Q[W_\theta f_\theta], \qquad \widehat{g}_N(\theta) := \frac{1}{N}\sum_{i=1}^{N} W_{\theta,i} f_{\theta,i}, \tag{2}$$

where $f_\theta := A(X, Y)\sum_{t=1}^{T} \ell_{\theta,t}$ denotes the score and advantage contribution and $W_\theta = \prod_{t=1}^{T} r_{\theta,t}$ becomes the complete sample likelihood ratio.

Equation (2) reveals the connection between policy-gradient optimization and the importance-weighted mean-estimation problem, which is a foundational problem in statistics. The quality of the resulting policy update therefore depends on how reliably this finite importance-weighted average estimates $g(\theta)$; the update thus inherits the challenges of importance-weighted estimation.

Estimator choices. Large or highly dispersed likelihood ratios can make gradient estimators statistically unstable. Practical policy-gradient algorithms therefore modify the coefficients of the empirical gradient to control this behavior. To represent these modifications in a common form, write $\mathbf{r}_\theta = (r_{\theta,1}, \ldots, r_{\theta,T})$ and let $u = (u_1, \ldots, u_T)$ denote a collection of coefficient rules. Define

$$\psi_\theta(u; Z) := A(Z) \sum_{t=1}^{T} u_t(\mathbf{r}_\theta, A(Z)) \ell_{\theta,t}(Z), \qquad \widehat{g}_N(u; \theta) := \frac{1}{N} \sum_{i=1}^{N} \psi_\theta(u; Z_i). \tag{3}$$

Setting $u_{\mathrm{IS},t} = r_{\theta,t}$ for $t = 1, \ldots, T$ corresponds to the conventional IS estimator. We consider two common estimator choices in this paper: the truncated importance sampling rule (TIS) [5] and PPO/GRPO clipping [12, 13]:

$$u_{\mathrm{T},t}(\mathbf{r}, A) := \min\{r_t, c\}, \qquad u_{\mathrm{P},t}(\mathbf{r}, A) := r_t \left[\mathbf{1}\{A \geq 0,\ r_t \leq 1 + \epsilon\} + \mathbf{1}\{A < 0,\ r_t \geq 1 - \epsilon\}\right],$$

TIS caps the corresponding ratio, which reduces the magnitude of a large importance coefficient while retaining the associated score contribution. On the other hand, PPO/GRPO clipping modifies the gradient in a qualitatively different way by an advantage-dependent masking rule. Once the corresponding clipping condition is active, the selected score contribution is removed from the gradient.

3.2 Importance-weight concentration and effective sample size

Despite the central role of finite-sample gradient reliability in policy optimization, relatively few works directly connect this reliability to the statistical properties of the underlying gradient estimator. From the estimator perspective developed above, however, this question can be studied through the theory of importance-weighted mean estimation, whose finite-sample behavior has been extensively studied in statistics. A key quantity in this literature is the dispersion of the likelihood weights, which can be characterized through normalized effective sample size (ESS) [3, 4]. In our population setting, normalized ESS can be written as [3, 4, 10, 14]:

$$\rho_\theta = \frac{1}{\mathbb{E}_Q[(W_\theta - 1)^2] + 1}. \tag{4}$$

A low ESS therefore indicates greater dispersion of the population likelihood ratios and suggests greater finite-sample variability in importance-weighted estimation. To illustrate, we use Figure 2 in the Gaussian policy optimization example in Section 5.1. Here, lower ESS worsens the gradient estimator MSE. Intuitively, this further suggests that ESS provides the desired connection to reliable policy updates in Figure 1. When ESS approaches the observed threshold, subsequent updates become unstable and eventually collapse.

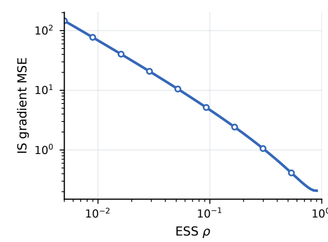

Figure 2: Exact IS gradient MSE as a function of ESS ρ in the Gaussian bandit.

4 Theoretical Analysis

In this section, we provide the theoretical analysis of the finite sample policy improvement and the estimator reliability, along with the connection with ESS.

4.1 Gradient-estimator reliability and policy improvement

MSE and policy improvement We first establish how mean squared error (MSE) in the estimated gradient limits one-step policy improvement. Under the smoothness condition on the parameterized policy, one can convert an arbitrary estimator's deviation from the population gradient directly into a lower bound on policy improvement.

Lemma 1 (A reliable gradient estimate yields policy improvement)**.** *Let $\widehat{g}$ be an estimate of $g = \nabla J(\theta)$ satisfying $\|\widehat{g} - g\|_2 \leq \varepsilon$. Suppose that J is L-smooth along the update segment for $L > 0$, and define*

$$\gamma := \frac{1}{L}\left(1 - \frac{\varepsilon}{\|\widehat{g}\|_2}\right)_+, \tag{5}$$

with $\gamma = 0$ when $\widehat{g} = 0$. Then

$$J(\theta + \gamma\widehat{g}) - J(\theta) \geq \frac{1}{2L}\left(\|\widehat{g}\|_2 - \varepsilon\right)_+^2. \tag{6}$$

Lemma 1 identifies the quantity that determines whether a finite-sample direction is useful: a positive certificate for policy optimization is obtained whenever the gradient magnitude exceeds its estimation-error radius. Note that in practice, such error radius is not known, hence requires a specific characterization that can be used in practice.

IS reliability through ESS. We show that the error can be characterized explicitly by ESS. Throughout, we condition on the preceding updates and write $W_k := W_{\theta_k}$, $f_k := f_{\theta_k}$, $g_k := g(\theta_k)$, $\widehat{g}_k := \widehat{g}_N(\theta_k)$, and $\rho_k := \rho_{\theta_k}$ at update k. The following Lemma provides the connection with a high-probability reliability radius and ESS.

Lemma 2 (ESS controls IS gradient reliability)**.** *Suppose $d_2(P_{\theta_k}\|Q) < \infty$ and $\mathbb{E}_Q[W_k^2\|f_k\|_2^2] < \infty$ and define $M_{2,k} := \frac{\mathbb{E}_Q[W_k^2\|f_k\|_2^2]}{\mathbb{E}_Q[W_k^2]}$. Then, for any $0 < \delta \leq 1$,*

$$\Pr\left(\|\widehat{g}_k - g_k\|_2 \leq \sqrt{\frac{M_{2,k}}{\delta N\rho_k}}\right) \geq 1 - \delta. \tag{7}$$

Thus the error radius entering Lemma 1 scales as $\rho_k^{-1/2}$. Substituting this radius gives the corresponding ESS-dependent improvement guarantee.

Theorem 3 (ESS-dependent improvement for the IS estimator)**.** *Under the conditions of Lemma 2, suppose that J is L_k-smooth along the update segment. For any $0 < \delta \leq 1$, define*

$$\varepsilon_k(\delta) := \sqrt{\frac{M_{2,k}}{\delta N\rho_k}}, \qquad \gamma_k := \frac{1}{L_k}\left(1 - \frac{\varepsilon_k(\delta)}{\|\widehat{g}_k\|_2}\right)_+, \tag{8}$$

with $\gamma_k = 0$ when $\widehat{g}_k = 0$. Then, with probability at least $1 - \delta$,

$$J(\theta_k + \gamma_k\widehat{g}_k) - J(\theta_k) \geq \frac{1}{2L_k}\left(\|\widehat{g}_k\|_2 - \varepsilon_k(\delta)\right)_+^2. \tag{9}$$

The preceding results establish the complete IS reliability chain. ESS enters the exact gradient MSE through the importance-weight second moment; this MSE determines a high-probability error radius, which in turn controls both the magnitude and the probability of one-step policy improvement.

4.2 Estimator reliability and ESS-dependent crossover

We now generalize the result to the general estimators. To facilitate the analysis, consider a general coefficient rule u. The same reliability question must also account for the bias introduced by modifying the estimator. Write the MSE error of the gradient estimator as $m_k(u)$. We first connect MSE to the gradient-estimator reliability.

Lemma 4 (MSE controls gradient-estimator reliability)**.** *For any coefficient rule u with $m_k(u) < \infty$ and any $0 < \delta \le 1$,*

$$\Pr\left(\|\widehat{g}_k(u) - g_k\|_2 \le \sqrt{\frac{m_k(u)}{\delta}} \right) \ge 1 - \delta. \tag{10}$$

Together with Lemma 1, Lemma 4 shows that smaller MSE yields a tighter high-probability policy-improvement guarantee. Estimator selection, however, is not equivalent to MSE minimization. A rule may reduce its MSE by suppressing gradient contributions that are themselves useful for policy improvement. The resulting optimization tradeoff is captured by the following certificate.

Theorem 5 (MSE form of the policy-improvement certificate)**.** *Suppose that J is L_k-smooth on every update segment generated by u almost surely and that $\mathbb{E}\|\widehat{g}_k(u)\|_2^2 < \infty$. For any $\eta_k > 0$,*

$$\mathbb{E}[J(\theta_k + \eta_k \widehat{g}_k(u)) - J(\theta_k)] \ge B_k(u; \eta_k), \tag{11}$$

where

$$B_k(u; \eta_k) = \frac{\eta_k}{2}\|g_k\|_2^2 - \frac{\eta_k}{2} m_k(u) + \frac{\eta_k}{2}(1 - L_k\eta_k)\mathbb{E}\|\widehat{g}_k(u)\|_2^2. \tag{12}$$

Equation (12) places estimator reliability and policy improvement in the same expression. Lower MSE strengthens the certificate, while coefficient control can simultaneously reduce the magnitude of the update. The preferred estimator therefore depends on whether the reduction in estimation error is large enough to compensate for the useful update removed by the modification. Such a preferred estimator can be determined by comparing the lower bounds on policy improvement.

Corollary 6 (Estimator crossover)**.** *Let u and v use the same step size $0 < \eta_k \le 1/L_k$. Then*

$$B_k(u; \eta_k) > B_k(v; \eta_k) \quad \Longleftrightarrow \quad m_k(v) - m_k(u) > (1 - L_k\eta_k)\left\{\mathbb{E}\|\widehat{g}_k(v)\|_2^2 - \mathbb{E}\|\widehat{g}_k(u)\|_2^2\right\}. \tag{13}$$

Corollary 6 shows that estimator selection is determined by a tradeoff between the reduction in estimation error and the corresponding change in update magnitude. Evaluating this crossover directly, however, requires population MSEs and second moments. The results of Section 4.1 allow its contribution to be expressed in terms of ESS, yielding the following specialization.

Proposition 7 (ESS-dependent estimator crossover)**.** *Let u be an alternative coefficient rule and suppose $0 < \eta_k \le 1/L_k$. Then u has a stronger certificate than IS if and only if*

$$\frac{L_k\eta_k M_{2,k}}{N\rho_k} > m_k(u) - (1 - L_k\eta_k)\mathbb{E}\|\widehat{g}_k(u)\|_2^2 + \left(1 - L_k\eta_k + \frac{L_k\eta_k}{N}\right)\|g_k\|_2^2. \tag{14}$$

Proposition 7 isolates the role of ESS in the crossover. All dependence of the IS certificate on importance-weight concentration appears through $1/\rho_k$ on the left-hand side, whereas the right-hand side collects the corresponding MSE and second-moment terms of the alternative estimator. Conditional on these quantities, decreasing ESS makes the inequality easier to satisfy and therefore shifts the comparison toward estimator control. The precise crossover remains estimator dependent, which allows truncation and clipping to become preferable at different ESS levels.

TIS under rare truncation. TIS changes the estimator only on ratios exceeding its cap. Let $u_{\mathrm{T}}^{(\infty)}$ denote the corresponding untruncated factorwise importance-weighted rule, $u_{\mathrm{T},t}^{(\infty)} = r_t$. The following rule demonstrates the similarity between TIS and IS estimators.

Proposition 8 (Truncation preserves the importance-weighted certificate)**.** *Suppose that the conditional second moment of $A(r_{\theta_k,t} - c)_+ \ell_{\theta_k,t}$ on $\{r_{\theta_k,t} > c\}$ is uniformly bounded, and write $p_{c,k} := \frac{1}{T}\sum_{t=1}^{T} \Pr(r_{\theta_k,t} > c)$. Then, for fixed remaining moments,*

$$\left| B_k(u_{\mathrm{T}}; \eta_k) - B_k(u_{\mathrm{T}}^{(\infty)}; \eta_k) \right| = O(\sqrt{p_{c,k}}) \qquad \text{as } p_{c,k} \to 0. \tag{15}$$

Proposition 8 shows that TIS retains the improvement certificate of untruncated importance weighting when truncation affects only a small, controlled tail. PPO-style clipping makes a stronger modification, which masks the entire r_t-weighted contribution. It can therefore produce a larger reduction in estimator variation, but requires that reduction to compensate for a correspondingly larger change in the mean update.

Together, our analysis reveals the importance of the policy gradient reliability in policy optimization and suggests ESS as the observable reliability criterion. Further, Theorem 3 and Proposition 7 provide insight into when other estimators can be beneficial in some cases. High ESS suggests using permissive gradient estimators, and lower ESS suggests more conservative choices such as clipping, highlighting the potential benefit of stronger estimator variance control.

5 Policy optimization and ESS

In this section, we demonstrate our theoretical analysis in two settings. Here, we present our results on the Gaussian bandit and LLM trajectories.

5.1 Policy optimization in a Gaussian bandit

We consider Gaussian policies with quadratic rewards, which admit closed-form population gradients and estimator moments. We first test whether ESS predicts when IS loses finite-sample reliability and then compare policy-improvement gains along a path of varying ESS. Appendix A gives the calculations and protocol.

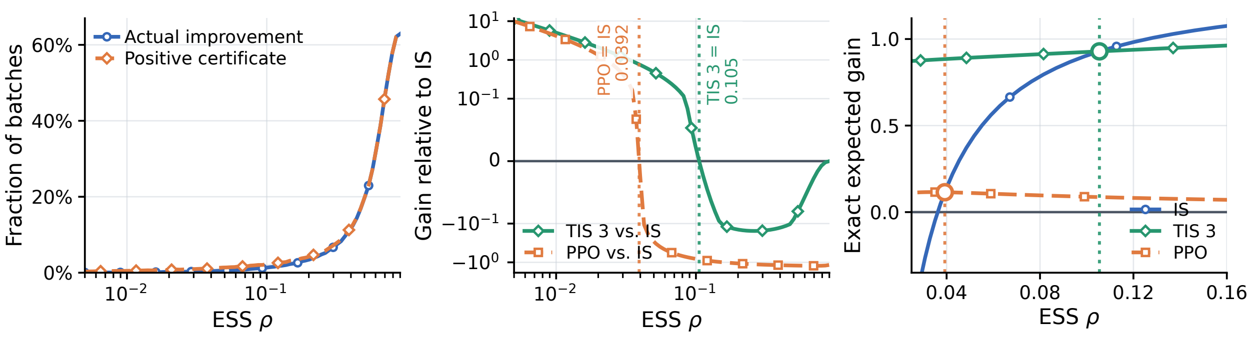

Figure 3: Gaussian-bandit reliability and estimator comparison. From left to right: positive certificates and actual positive improvement under the adaptive IS step; exact expected gain of TIS-3 and PPO relative to IS; and exact expected gain near the pairwise crossovers. Vertical lines and open markers identify the intersections.

ESS predicts reliability and policy improvement The left plot in Figure 3 illustrates how estimator reliability translates into policy improvement. Across the full ESS path, the probability of actual improvement tracks the probability of a positive certificate with a maximum gap of only 0.00891. As support contracts, both probabilities fall together. ESS therefore identifies when a reliable IS update is available, but this first stage leaves one question open: which controlled estimator should be used after IS loses its advantage?

ESS determines estimator-choice criterion Figure 3 then demonstrates the estimator-selection criterion in Corollary 6. The center plot gives the resulting expected-gain difference relative to IS. TIS-3 has the lowest MSE throughout the path, yet IS retains more useful gradient alignment and therefore has the larger gain at high ESS. As ESS falls, the variance reduction from truncation becomes decisive, and TIS-3 overtakes IS at $\rho = 0.105$. PPO removes still more alignment, so it crosses IS only at $\rho = 0.0392$. The right plot makes both transitions visible as intersections of the exact policy-improvement curves. The resulting upper envelope selects IS in the high-support regime and TIS-3 in the low-support regime, exactly as predicted by the gain criterion.

5.2 Policy optimization in large language models

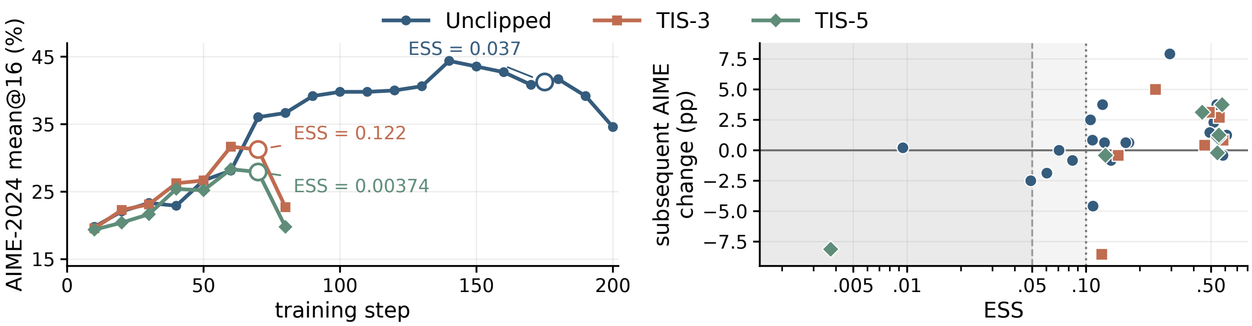

Figure 4: Left: AIME-2024 mean@16 along an unclipped trajectory and TIS with caps 3 and 5. Annotations report ESS at the onset of the sustained performance decline. Right: each point pairs ESS at an evaluation checkpoint with the change in AIME-2024 mean@16 over the following ten updates. The shaded regions indicate ESS below 0.1 and 0.05. Lower ESS is associated with substantially weaker and less reliable subsequent improvement.

We now examine whether the same finite-sample mechanism is visible during large-model optimization. We consider three Qwen3-30B-A3B trajectories, consisting of an unclipped update and TIS with caps 3 and 5, and evaluate all trajectories on AIME-2024 using mean@16.

The left panel of Figure 4 places ESS within the training dynamics. All three importance-weighted updates improve substantially during the early stage of training, but their performance eventually begins to degrade. At the onset of the sustained decline, all three methods exhibit low ESS. The right panel tests the theoretical prediction more directly. At each evaluation checkpoint, we pair the current ESS with the change in AIME-2024 mean@16 over the following ten updates. The resulting pattern shows a clear change in update reliability as ESS decreases. Low ESS identifies a regime in which continued importance-weighted updates are substantially less likely to produce reliable improvement.

This behavior is the effect characterized in Section 4. Decreasing ESS enlarges the gradient-error radius through the factor $\rho_k^{-1/2}$ and consequently weakens the corresponding policy-improvement certificate. The large-model trajectories exhibit the same optimization consequence: as ESS deteriorates, subsequent improvement becomes less reliable. This observation motivates the optimization rule studied next: retain the importance-weighted update while its reliability remains high, and introduce clipping when ESS indicates that additional estimator control has become valuable.

6 ESS-conditioned policy optimization

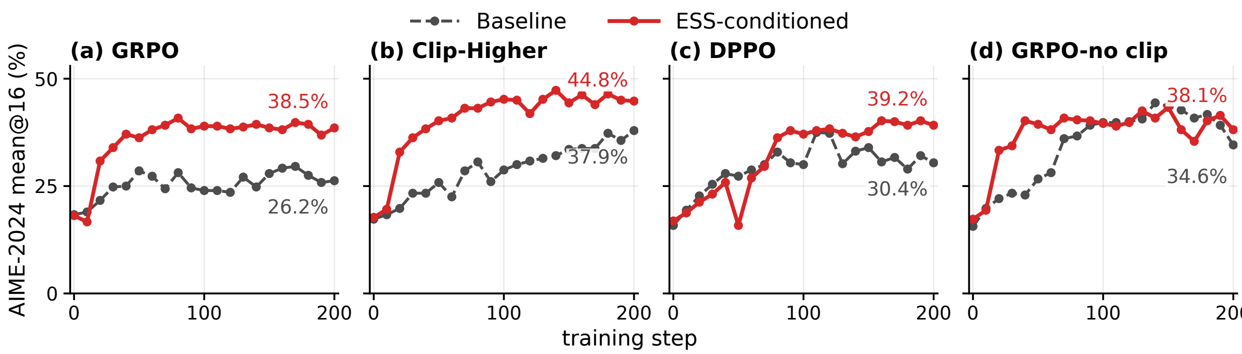

Figure 5: Qwen3-30B-A3B math RLVR results. Dashed curves correspond to the baseline methods. Solid curves begin with TIS-3 in (a), (b), and (c), or an unclipped update in (d), and switch to the corresponding clipped update when ESS falls below 0.1. (a) GRPO clipping. (b) Clip-Higher. (c) DPPO. (d) GRPO-no clip.

In this section, we now extend our theoretical analysis to an optimization rule. Based on this observation, the reliability signal can be used in optimization to determine the *clipping timing*. While ESS remains above a fixed threshold, we retain importance-weighted updates and activate clipping for conservative updates.

6.1 Update rule and protocol

Let $\widehat{g}_{\mathrm{IW}}$ denote the importance-weighted update such as IS or TIS, and $\widehat{g}_{\mathrm{clip}}$ a prespecified clipped update. Given an ESS threshold τ, we use the following gradient estimator:

$$\widehat{g}_{\mathrm{ESS}} = \begin{cases} \widehat{g}_{\mathrm{IW}}, & \widehat{\rho} \geq \tau, \\ \widehat{g}_{\mathrm{clip}}, & \widehat{\rho} < \tau. \end{cases} \tag{16}$$

We use $\tau = 0.1$ in the primary experiments.

In the main experiments, we investigate the configurations that demonstrated failure modes, and see whether our intervention rule yields reliable updates. We consider the following on Qwen3-30B-A3B base models: (1) TIS-3 updates with GRPO clipping [13], (2) TIS-3 updates with Clip-Higher [15], (3) TIS-3 updates with DPPO [9], (4) unclipped updates with GRPO clipping [13]. Baseline models include GRPO, GRPO with no clips, Clip-Higher, DPPO, and TIS-3 updates. Our main experiments train models for up to 200 training steps on the mathematical-reasoning data of Yu et al. [15] and evaluate AIME-2024 using mean@16.

6.2 Results

The first three panels of Figure 5 compare ESS-conditioned optimization with static clipping across corresponding baselines. In all three cases, ESS-conditioned optimization presents stronger AIME performance and efficiency. Specifically, while TIS-3 and unclipped GRPO present instability, the ESS-conditioned optimization exhibits remarkable stability, preventing catastrophic collapses.

Figure 5(d) leaves one run unchanged and introduces conventional clipping in the other when ESS falls below 0.1. While the unclipped run subsequently degrades to 34.6%, ESS-conditioned learning retains 38.1% at step 200. The timing principle therefore extends beyond the TIS-3 branch.

Together, these panels show the empirical strength of the ESS-conditioned optimization, which supports the theoretical motivation.

6.3 Broader Evaluation

Results on Different Scales. We first change model scale while retaining the mathematical-reasoning objective. Figure 6 extends the recipe comparison to dense Qwen3-8B. ESS-conditioned optimization demonstrates better sample efficiency than its clipped counterpart under both standard GRPO and Clip-Higher.

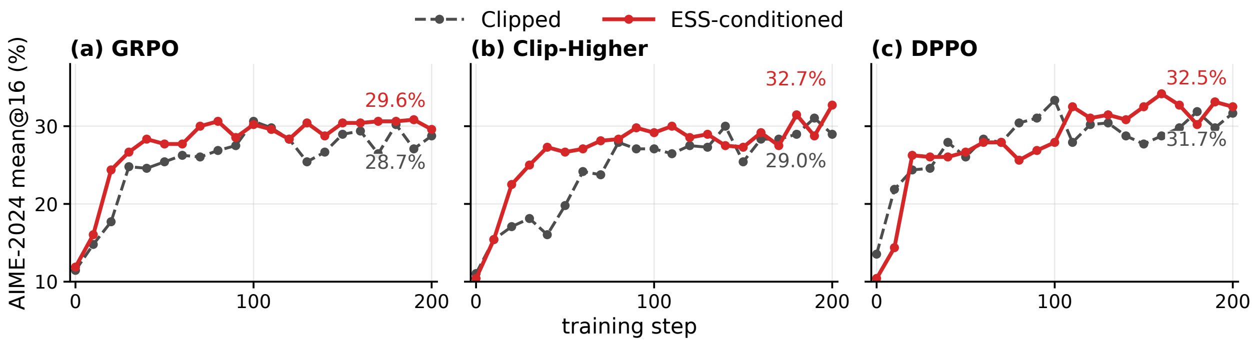

Figure 6: Qwen3-8B math RLVR over 200 policy updates. Dashed curves apply clipping throughout training, and solid curves use the ESS-conditioned counterpart. (a) GRPO. (b) Clip-Higher. (c) DPPO. End labels report the step-200 AIME-2024 mean@16.

We then reduce the model to Qwen3-1.7B while retaining the mathematical-reasoning objective. Figure **??** reports evaluation performance for the two matched learning-rate recipes and overlays ESS for TIS-3 + Clip. ESS remains above the conditioning threshold of 0.1, so the optimizer never activates clipping and thus behaves like TIS-3. Performance remains stable and improves even during the extended 500-step training run. This theoretical explanation is consistent with observations in other work, where TIS shows reasonable training trajectories.

Results on RLHF. We further change both the objective and the data by training Qwen3-4B-Instruct-2507 on Anthropic HH-RLHF [1], using Skywork-Reward-Llama-3.1-8B [2] as the reward model. Figure 7 shows the same advantage under learned-reward optimization. The TIS-3-plus-ESS recipe reaches a final reward-model score of 60.7, compared with 43.0 for GRPO. This experiment evaluates optimization of the specified reward model rather than human preference directly.

6.4 Ablations

The ESS-conditioned construction contains two distinct choices: the threshold determines when the update changes, while the low-ESS rule determines how it changes. We study these components separately.

ESS threshold. Figure 8 varies only τ in the TIS-3/GRPO construction. Thresholds 0.05, 0.1, and 0.2 all avoid the collapse of ungated TIS-3, with last-observed AIME scores of 43.1%, 38.5%, and 34.8%, respectively, demonstrating that the collapse-prevention behavior persists across the tested range.

Clipping versus skipping. ESS determines when to intervene but does not prescribe the intervention itself. We therefore replace clipping with a complete update veto while keeping the same ESS threshold.

Figure 9 isolates a complete update veto on the unclipped branch. Skipping raises the final score only slightly, from 34.6% to 35.6%, while reducing the peak from 44.4% to 35.6%. By contrast, ESS-conditioned clipping in Figure 5(d) reaches a 43.3% peak and retains 38.1%. The intervention must therefore preserve useful gradient signal after ESS identifies an unreliable minibatch.

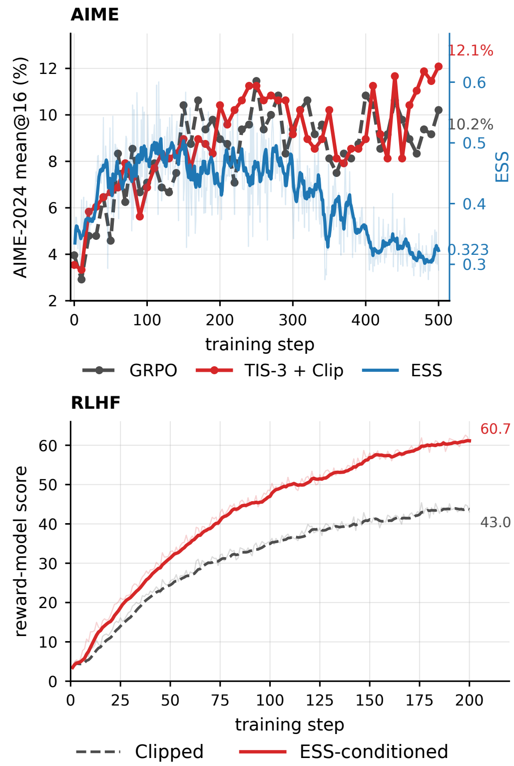

Figure 7: Top: Qwen3-1.7B results over extended training. AIME-2024 result is shown for GRPO (dashed) and TIS-3 + Clip (solid). The blue line represents ESS for TIS-3 + Clip. Bottom: RLHF on Qwen3-4B-Instruct-2507. End labels report the step-200 observations.

7 Conclusion

We studied a finite-sample perspective on policy optimization that provides a theoretical foundation for finite-sample policy improvement. Off-policy gradients admit an importance-weighted mean-estimation formulation, under which ESS enters directly through the second moment of the importance weights. Our theory connects this finite-sample estimation error to policy improvement and separately characterizes when clipping compensates for the gradient alignment it removes. We demonstrated our theoretical analysis through two empirical analyses, one in a Gaussian bandit and one in large language models. We then extended our analysis to a simple reliability-aware optimization method, which prevents the collapse seen with the previous updates and demonstrates strong empirical performance across model sizes and objectives.

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A Details for Gaussian Bandit Experiments

This section provides the experimental setup and exact calculations underlying the Gaussian bandit experiments in Section 5.1. All estimator moments, mean-squared errors, and expected one-step gains are computed analytically using truncated Gaussian moments. Monte Carlo simulation is used only to evaluate the batch-level probabilities in Figure 3.

A.1 Experimental setup

Policy pair and ESS. We fix the current policy and consider a family of rollout policies

$$P_\theta = \mathcal{N}(\theta, 1), \qquad Q_\Delta = \mathcal{N}(\theta - \Delta, 1), \qquad \Delta \geq 0.$$

Thus, Δ controls only the distance between the current and rollout policies. Defining the centered action $z := a - \theta$, we have

$$z \sim \mathcal{N}(0, 1) \quad \text{under } P_\theta, \qquad z \sim \mathcal{N}(-\Delta, 1) \quad \text{under } Q_\Delta.$$

The corresponding likelihood ratio is

$$W_\Delta(z) = \frac{dP_\theta}{dQ_\Delta}(z) = \exp\left\{\Delta z + \frac{\Delta^2}{2}\right\}.$$

A direct calculation gives

$$\mathbb{E}_{Q_\Delta}[W_\Delta^2] = e^{\Delta^2}, \qquad \rho(\Delta) = e^{-\Delta^2}, \qquad \Delta = \sqrt{-\log \rho}. \tag{17}$$

Hence, Δ and ESS are in one-to-one correspondence. We therefore parameterize all figures by ρ, while carrying out the calculations using the corresponding value of Δ.

Reward and policy-gradient contribution. We use the quadratic reward

$$R(a) = -\frac{1}{2}(a - a^\star)^2$$

and define $d := a^\star - \theta > 0$. The current-policy objective and population gradient are

$$J(\theta) = -\frac{1}{2}(d^2 + 1), \qquad \nabla J(\theta) = d.$$

Throughout the experiment, we hold θ, and therefore d, fixed while varying Δ.

Using the exact current-policy value as the baseline, the advantage for a centered action z is

$$A(z) = dz + \frac{1 - z^2}{2}.$$

Since the score of $\mathcal{N}(\theta, 1)$ is z, the corresponding per-sample policy-gradient contribution is

$$\psi(z) := zA(z) = dz^2 + \frac{z - z^3}{2}, \qquad \mathbb{E}_{P_\theta}[\psi(z)] = d.$$

Estimators. Given samples

$$z_1, \dots, z_N \overset{\text{iid}}{\sim} \mathcal{N}(-\Delta, 1),$$

we compare IS, TIS-3, and PPO. Their per-sample contributions are

$$\begin{aligned}
\xi_{\mathrm{IS}}(z) &:= W_\Delta(z)\psi(z), \\
\xi_{\mathrm{T}}(z) &:= \min\{W_\Delta(z), c\}\psi(z), \\
\xi_{\mathrm{P}}(z) &:= W_\Delta(z)\psi(z)M_{\epsilon,\Delta}(z),
\end{aligned}$$

where $c = 3$, $\epsilon = 0.2$, and

$$M_{\epsilon,\Delta}(z) = \mathbf{1}\{A(z) \geq 0,\ W_\Delta(z) \leq 1+\epsilon\} + \mathbf{1}\{A(z) < 0,\ W_\Delta(z) \geq 1-\epsilon\}.$$

The corresponding batch estimator is

$$\widehat{g}_u = \frac{1}{N}\sum_{i=1}^{N} \xi_u(z_i), \qquad u \in \{\mathrm{IS}, \mathrm{T}, \mathrm{P}\}.$$

A.2 Exact calculation of evaluation quantities

All quantities reported in the Gaussian experiments, except for the batch-level probabilities, are computed exactly. For each estimator, define its first two per-sample moments by

$$\mu_u(\Delta) := \mathbb{E}_{Q_\Delta}[\xi_u(z)], \qquad \nu_u(\Delta) := \mathbb{E}_{Q_\Delta}[\xi_u(z)^2].$$

The corresponding batch moments and mean-squared error are

$$\begin{aligned}
\mathbb{E}[\widehat{g}_u] &= \mu_u, \\
\mathbb{E}[\widehat{g}_u^2] &= \mu_u^2 + \frac{\nu_u - \mu_u^2}{N}, \\
\mathbb{E}[(\widehat{g}_u - d)^2] &= (\mu_u - d)^2 + \frac{\nu_u - \mu_u^2}{N}.
\end{aligned}$$

Thus, the pair (μ_u, ν_u) determines both the gradient-estimation MSE and the second-moment term appearing in the policy-improvement certificate.

For IS, change of measure gives

$$\mu_{\mathrm{IS}} = d, \qquad \nu_{\mathrm{IS}} = e^{\Delta^2}\mathbb{E}_{V\sim\mathcal{N}(\Delta,1)}[\psi(V)^2].$$

For PPO,

$$\begin{aligned}
\mu_{\mathrm{P}} &= \mathbb{E}_{U\sim\mathcal{N}(0,1)}[\psi(U)M_{\epsilon,\Delta}(U)], \\
\nu_{\mathrm{P}} &= e^{\Delta^2}\mathbb{E}_{V\sim\mathcal{N}(\Delta,1)}[\psi(V)^2 M_{\epsilon,\Delta}(V)].
\end{aligned}$$

For TIS-3, the truncation boundary obtained from $W_\Delta(z) = c$ is

$$b_c(\Delta) := \frac{\log c - \Delta^2/2}{\Delta}.$$

Splitting the integral at this boundary yields

$$\begin{aligned}
\mu_{\mathrm{T}} &= \mathbb{E}_{U\sim\mathcal{N}(0,1)}[\psi(U)\mathbf{1}\{U \leq b_c\}] + c\,\mathbb{E}_{U\sim\mathcal{N}(-\Delta,1)}[\psi(U)\mathbf{1}\{U > b_c\}], \\
\nu_{\mathrm{T}} &= e^{\Delta^2}\mathbb{E}_{V\sim\mathcal{N}(\Delta,1)}[\psi(V)^2\mathbf{1}\{V \leq b_c\}] + c^2\mathbb{E}_{V\sim\mathcal{N}(-\Delta,1)}[\psi(V)^2\mathbf{1}\{V > b_c\}].
\end{aligned}$$

These expectations have closed forms because ψ is a cubic polynomial and all estimator boundaries are linear thresholds in z. In particular, the advantage changes sign at

$$z = d \pm \sqrt{d^2 + 1},$$

while the PPO ratio boundaries are

$$b_\pm(\Delta) = \frac{\log(1 \pm \epsilon) - \Delta^2/2}{\Delta}.$$

The real line can therefore be partitioned at the two advantage roots and the two PPO ratio boundaries, with the mask remaining constant on each resulting interval.

For completeness, define the truncated standard-normal moments

$$I_j(a, b) := \int_a^b z^j \phi(z)\, dz.$$

They satisfy

$$I_0(a, b) = \Phi(b) - \Phi(a), \qquad I_1(a, b) = \phi(a) - \phi(b),$$

and, for $j \geq 2$,

$$I_j(a, b) = a^{j-1}\phi(a) - b^{j-1}\phi(b) + (j-1)I_{j-2}(a, b).$$

After shifting the integration variable for nonzero Gaussian means, all moments above reduce to finite sums of these truncated Gaussian moments.

A.3 ESS and IS reliability

Figures 2 and 3 evaluate the ESS-dependent reliability result for the IS estimator. We set

$$N = 32, \qquad d = 2, \qquad \delta = 0.1.$$

Exact MSE and theoretical envelope. Define

$$M_2(\Delta) := \rho(\Delta)\nu_{\mathrm{IS}}(\Delta) = \mathbb{E}_{V\sim\mathcal{N}(\Delta,1)}[\psi(V)^2].$$

For $d = 2$, this quantity is

$$M_2(\Delta) = \frac{\Delta^6}{4} - 2\Delta^5 + \frac{29\Delta^4}{4} - 18\Delta^3 + \frac{65\Delta^2}{2} - 24\Delta + \frac{29}{2}.$$

Since IS is unbiased, its exact MSE and the finite-moment upper bound in Equation (**??**) are

$$\mathrm{MSE}_{\mathrm{IS}}(\Delta) = \frac{1}{N}\left\{\frac{M_2(\Delta)}{\rho(\Delta)} - d^2\right\},$$
$$\mathrm{MSE}_{\mathrm{IS}}(\Delta) \leq \frac{M_2(\Delta)}{N\rho(\Delta)}.$$

Figure 2 plots the exact MSE against $\rho = e^{-\Delta^2}$. No Monte Carlo approximation is used in this figure.

Batch-level improvement experiment. The left plot in Figure 3 evaluates the corresponding batch-level policy-improvement statement. We consider 81 ESS values logarithmically spaced between 0.005 and 0.9. For each ρ, we set $\Delta = \sqrt{-\log \rho}$ and compute the theoretical error radius

$$\varepsilon(\rho) = \sqrt{\frac{M_2(\Delta)}{\delta N \rho}}.$$

At each ESS value, we independently generate 10^5 batches of size N from Q_Δ and compute

$$\widehat{g}_{\mathrm{IS}} = \frac{1}{N}\sum_{i=1}^{N} W_\Delta(z_i)\psi(z_i).$$

We then use the adaptive step from Theorem 3,

$$\gamma = \left(1 - \frac{\varepsilon(\rho)}{|\widehat{g}_{\mathrm{IS}}|}\right)_+,$$

where $L = 1$. The theorem gives a positive certificate whenever

$$|\widehat{g}_{\mathrm{IS}}| > \varepsilon(\rho).$$

For the same batch, the exact change in the quadratic objective is

$$\Delta J = d\gamma\widehat{g}_{\mathrm{IS}} - \frac{\gamma^2}{2}\widehat{g}_{\mathrm{IS}}^2.$$

The figure reports the fraction of batches satisfying the positive-certificate condition and the fraction satisfying $\Delta J > 0$. These correspond to the "positive certificate" and "actual improvement" curves, respectively.

As an additional check of the high-probability result, we record the event

$$|\widehat{g}_{\mathrm{IS}} - d| \le \varepsilon(\rho).$$

Across the displayed ESS grid, the minimum empirical coverage is 0.98277, exceeding the required level $1 - \delta = 0.9$.

A.4 Estimator crossover

The center and right plots in Figure 3 compare IS, TIS-3, and PPO under the common fixed step size $\eta = 0.4$. For the quadratic objective, the smoothness inequality is exact: for any realized scalar update h,

$$J(\theta + \eta h) - J(\theta) = \eta d h - \frac{\eta^2}{2}h^2.$$

Taking expectation with $h = \widehat{g}_u$ gives

$$\mathcal{G}_u(\eta) := \mathbb{E}[J(\theta + \eta\widehat{g}_u) - J(\theta)] = \eta d\mu_u - \frac{\eta^2}{2}\left(\mu_u^2 + \frac{\nu_u - \mu_u^2}{N}\right). \tag{18}$$

Hence, the fixed-step certificate in Theorem 5 coincides with the exact expected one-step improvement in this experiment.

Crossover criterion. The center plot shows

$$\mathcal{G}_{\mathrm{T}}(\eta) - \mathcal{G}_{\mathrm{IS}}(\eta) \qquad \text{and} \qquad \mathcal{G}_{\mathrm{P}}(\eta) - \mathcal{G}_{\mathrm{IS}}(\eta)$$

against ESS. Positive values indicate that the corresponding modified estimator yields larger expected one-step improvement than IS.

These gain differences are exactly equivalent to the crossover criterion in Corollary 6. Define

$$S_u := \mathbb{E}[\widehat{g}_u^2], \qquad m_u := \mathbb{E}[(\widehat{g}_u - d)^2].$$

Since $L = 1$,

$$\{m_{\mathrm{IS}} - m_u\} - (1-\eta)\{S_{\mathrm{IS}} - S_u\} = \frac{2}{\eta}\left\{\mathcal{G}_u(\eta) - \mathcal{G}_{\mathrm{IS}}(\eta)\right\}. \tag{19}$$

Therefore, the zero crossings in the center plot of Figure 3 are precisely the estimator crossovers predicted by the corollary.

Using the exact moments above, the resulting crossover ESS values are

$$\rho_{\mathrm{IS,TIS3}} = 0.105, \qquad \rho_{\mathrm{IS,PPO}} = 0.0392.$$

The right plot gives the exact gains $\mathcal{G}_{\mathrm{IS}}$, $\mathcal{G}_{\mathrm{T}}$, and $\mathcal{G}_{\mathrm{P}}$ in the neighborhood of these two crossover points. The vertical lines and open markers indicate the corresponding roots.

TIS-3 has the smallest gradient MSE throughout the displayed ESS range, but this does not make it the preferred update everywhere. At high ESS, IS retains the larger expected gain. TIS-3 overtakes IS only once ρ falls below 0.105, while PPO crosses IS later at $\rho = 0.0392$ and remains below TIS-3 throughout the displayed range. The experiment therefore illustrates the distinction established by the theory: estimator accuracy and policy improvement need not induce the same ordering.

B Details for LLM optimizations

In this section, we provide additional details on the experimental setup and report supplementary ablation studies.

B.1 Experimental Details

Most experiments are conducted using NVIDIA A100 and B200 GPUs. Our implementation is built on VeRL [**?**], with Megatron [**?**] used as the training backend and vLLM [**?**] used for efficient rollout generation. The main training configurations are summarized in Table 1.

For evaluation, we periodically assess each method throughout RL training on AIME24. We run evaluations every 10 training steps for all model scales. To ensure consistent comparisons across experiments, we use the same sampling configuration throughout, with temperature 0.7 and top-p 0.95.

B.2 Additional Results

Results on other benchmarks We show the results on other benchmark datasets.

ESS threshold. Figure 8 varies only τ in the TIS-3/GRPO construction. Thresholds 0.05, 0.1, and 0.2 all avoid the collapse of ungated TIS-3, with last-observed AIME scores of 43.1%, 38.5%, and 34.8%, respectively. The threshold therefore controls the timing of clipping, and the resulting performance varies with how long TIS-3 remains active. The collapse-prevention behavior persists across the tested range.

Table 1: Hyperparameters used in our experiments.

Hyperparameters	**Qwen3-8B-Base**	**Qwen3-30B-A3B-Base**	**Qwen3-1.7B-Base**
Learning Rate	10^{-6}	10^{-6}	10^{-6}
PPO Epochs	1	1	1
Max Prompt Length	2048	2048	2048
Max Response Length	8192	8192	8192
Train Batch Size	256	256	64
PPO Mini Batch Size	32	32	16
Rollout Temperature	1.0	1.0	1.0

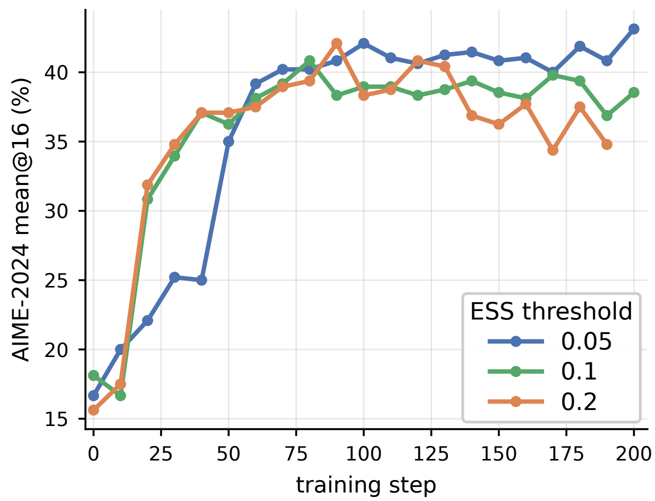

Figure 8: Sensitivity to the ESS threshold for TIS-3 with GRPO clipping on Qwen3-30B-A3B. Curves use $\tau \in \{0.05, 0.1, 0.2\}$; clipping is activated when ESS falls below the corresponding threshold.

Clipping versus skipping. ESS determines when to intervene but does not prescribe the intervention itself. We therefore replace clipping with a complete update veto while keeping the same ESS threshold.

Figure 9 isolates a complete update veto on the unclipped branch. Skipping raises the final score only slightly, from 34.6% to 35.6%, while reducing the peak from 44.4% to 35.6%. By contrast, ESS-conditioned clipping in Figure 5 reaches a 43.3% peak and retains 38.1%. The intervention must therefore preserve useful gradient signal after ESS identifies an unreliable minibatch.

The separate TIS-5 comparison in Figure 11 shows the complementary regime. Here skipping prevents complete collapse and raises the final AIME score from 0.00% to 39.4%. Together, the two comparisons separate the timing decision supplied by ESS from the estimator choice made after that transition.

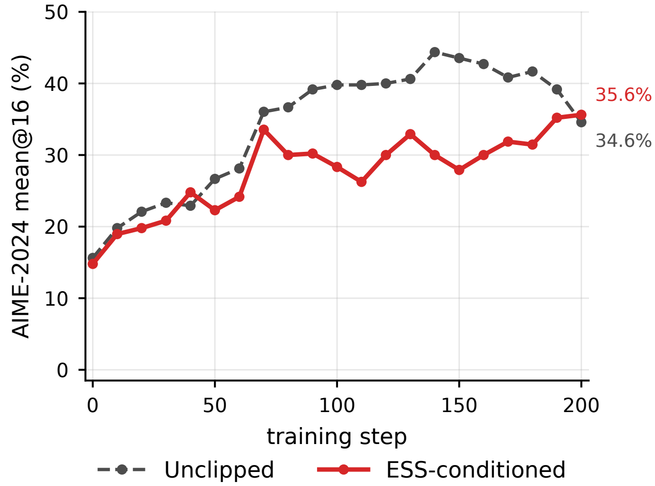

Figure 9: ESS-conditioned update skipping on the unclipped Qwen3-30B-A3B trajectory. The dashed curve applies every update, and the solid curve skips the update when ESS falls below 0.1. End labels report the final observed AIME-2024 mean@16.

Together, the ablations separate the two roles in ESS-conditioned optimization. The ESS threshold determines the intervention time; the post-threshold estimator determines the resulting bias–variation tradeoff. This distinction mirrors the theoretical analysis: identifying deterioration in importance-weighted reliability does not by itself determine how the gradient estimator should be modified.

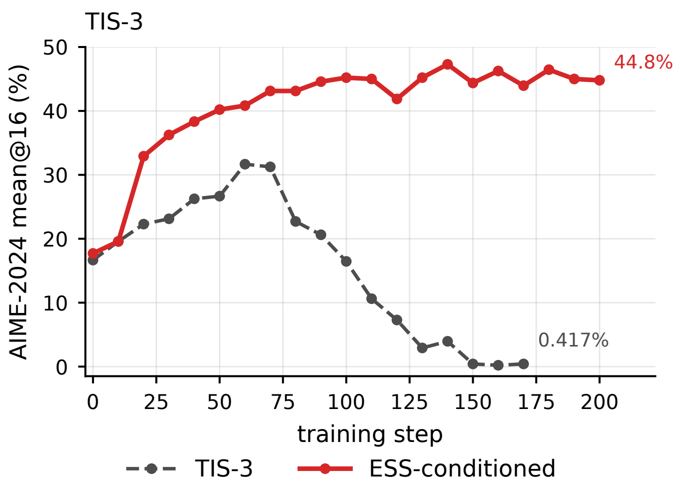

Figure 10: Separated TIS-3 collapse comparison from the main 30B experiment. Ungated TIS-3 is shown against the ESS-conditioned Clip-Higher trajectory used in Figure 5(b).

B.3 Additional results on training dynamics

The main text reports evaluation performance. We record here how often the conditional rule intervenes and the accompanying training dynamics.

Figure 12 shows the intended temporal ordering: intervention is already possible early, but its

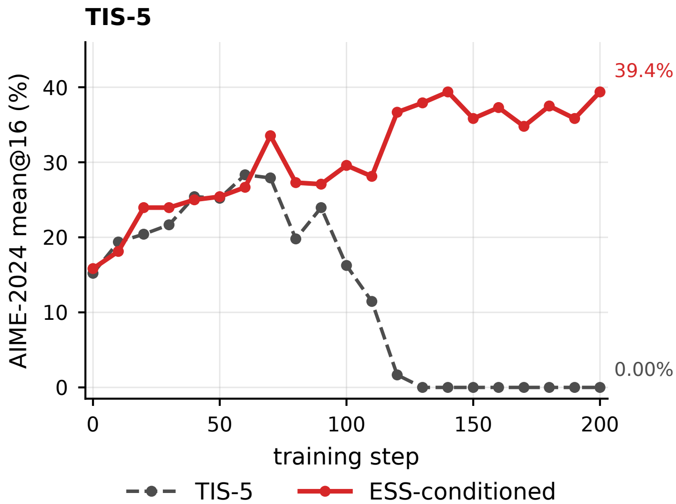

Figure 11: Separated TIS-5 skipping comparison on Qwen3-30B-A3B. The dashed curve continues TIS-5 throughout training; the solid curve skips low-ESS updates.

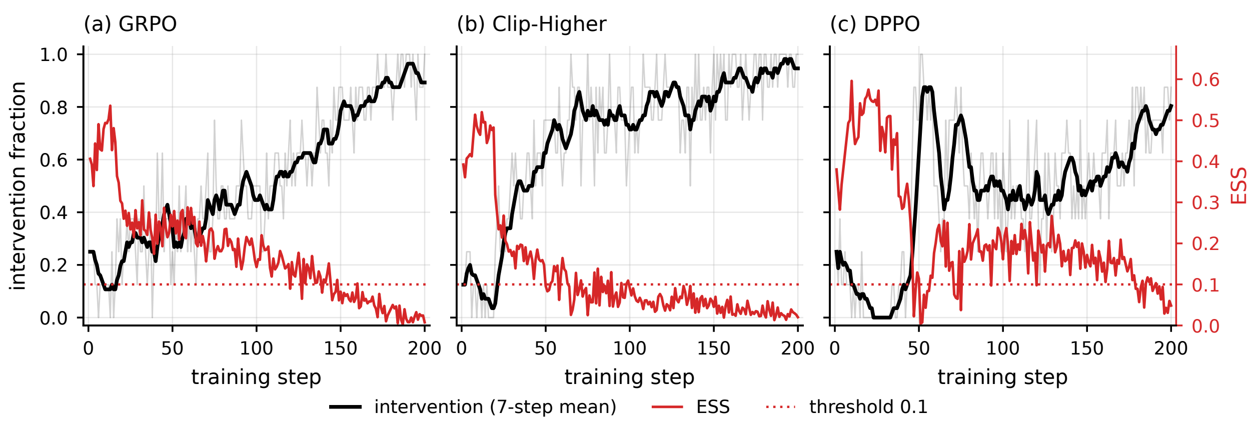

Figure 12: Realized ESS-conditioned intervention schedules. Each panel pairs the per-step intervention fraction (thin black) and its trailing seven-step mean (thick black) with ESS (red). GRPO and DPPO use raw ESS; Clip-Higher uses shaped ESS after the coefficient cap. The dotted line marks the threshold 0.1.

frequency increases as ESS contracts. The remaining figures record response length, entropy, and reward diagnostics.

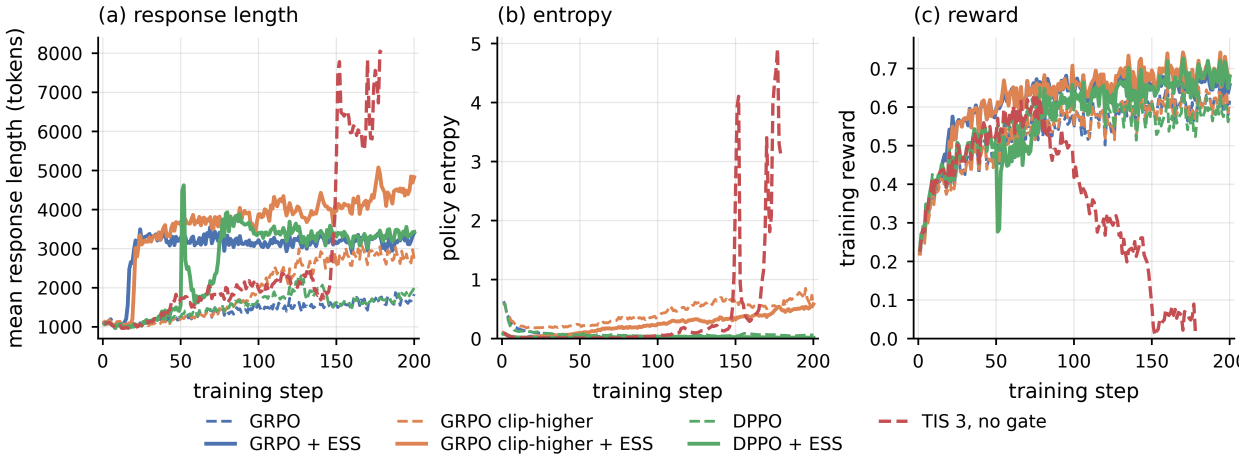

Figure 13: Training dynamics for the main comparisons. The static TIS 3 run develops extreme response length and entropy together with a collapse in training reward. The safeguards and their ESS-conditional variants avoid this failure over the reported 200 steps.

C Proofs for the theoretical results

At update k, all expectations and probabilities below are conditional on the preceding updates. The current policy is therefore fixed, and the rollout samples used at update k are independent draws from Q. We suppress this conditioning throughout.

To separate likelihood-ratio dispersion from the scale of the associated gradient contributions, define

$$M_{2,k} := \frac{\mathbb{E}_Q[W_k^2 \|f_k\|_2^2]}{\mathbb{E}_Q[W_k^2]} = \rho_k \mathbb{E}_Q[W_k^2 \|f_k\|_2^2]. \tag{20}$$

Since $\mathbb{E}_Q[W_k^2] = \rho_k^{-1}$ and $\mathbb{E}[\widehat{g}_k] = g_k$, the exact IS MSE is

$$\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 = \frac{1}{N}\left(\frac{M_{2,k}}{\rho_k} - \|g_k\|_2^2\right) \le \frac{M_{2,k}}{N\rho_k}. \tag{21}$$

Equation (21) isolates the role of ESS in the finite-sample error. The factor ρ_k^{-1} captures concentration of the importance weights, while $M_{2,k}$ captures the magnitude of the gradient contributions under the corresponding squared-weight tilt. For fixed $M_{2,k}$, the IS MSE therefore increases monotonically with ρ_k^{-1}. If f_k is bounded, then $M_{2,k} \le \|f_k\|_\infty^2$, recovering the standard bounded-contribution specialization.

C.1 IS identity and exact mean-square error

We first verify the unbiasedness and mean-square calculation used in Section 4.1. Recall that

$$f_k = (R - b(X))\nabla_\theta \log P_{\theta_k}(Y \mid X), \qquad W_k = \frac{dP_{\theta_k}}{dQ}.$$

By change of measure,

$$\mathbb{E}_Q[W_k f_k] = \mathbb{E}_{P_{\theta_k}}\left[(R - b(X))\nabla_\theta \log P_{\theta_k}(Y \mid X)\right].$$

The baseline term vanishes after conditioning on X, since

$$\begin{aligned}
&\mathbb{E}_{P_{\theta_k}}\left[b(X)\nabla_\theta \log P_{\theta_k}(Y \mid X) \mid X\right] \\
&\qquad = b(X) \int P_{\theta_k}(y \mid X)\nabla_\theta \log P_{\theta_k}(y \mid X)\, d\lambda(y) \\
&\qquad = b(X)\nabla_\theta \int P_{\theta_k}(y \mid X)\, d\lambda(y) = 0.
\end{aligned}$$

The remaining term is the score-function identity, and hence

$$\mathbb{E}_Q[W_k f_k] = \nabla J(\theta_k) = g_k.$$

Consequently, $\mathbb{E}[\widehat{g}_k] = g_k$.

To compute the MSE, let

$$\xi_i := W_{k,i} f_{k,i} - g_k.$$

Then $\mathbb{E}[\xi_i] = 0$ and

$$\widehat{g}_k - g_k = \frac{1}{N}\sum_{i=1}^{N} \xi_i.$$

Independence eliminates the cross terms, giving

$$\begin{aligned}
\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 &= \frac{1}{N}\mathbb{E}_Q\|W_k f_k - g_k\|_2^2 \\
&= \frac{1}{N}\left\{\mathbb{E}_Q[W_k^2\|f_k\|_2^2] - \|g_k\|_2^2\right\}.
\end{aligned}$$

Since

$$\mathbb{E}_Q[W_k^2] = \rho_k^{-1}, \qquad M_{2,k} = \rho_k \mathbb{E}_Q[W_k^2\|f_k\|_2^2],$$

we obtain

$$\mathbb{E}\|\widehat{g}_k - g_k\|_2^2 = \frac{1}{N}\left(\frac{M_{2,k}}{\rho_k} - \|g_k\|_2^2\right) \le \frac{M_{2,k}}{N\rho_k},$$

which proves Equation (21).

C.2 Proof of Lemma 1

Suppose $\|\widehat{g} - g\|_2 \le \varepsilon$. By L-smoothness, for every $\gamma \ge 0$,

$$J(\theta + \gamma\widehat{g}) - J(\theta) \ge \gamma g^\top \widehat{g} - \frac{L\gamma^2}{2}\|\widehat{g}\|_2^2.$$

Moreover,

$$\begin{aligned}
g^\top \widehat{g} &= \|\widehat{g}\|_2^2 + (g - \widehat{g})^\top \widehat{g} \\
&\ge \|\widehat{g}\|_2^2 - \|g - \widehat{g}\|_2\|\widehat{g}\|_2 \\
&\ge \|\widehat{g}\|_2\left(\|\widehat{g}\|_2 - \varepsilon\right).
\end{aligned}$$

Therefore

$$J(\theta + \gamma\widehat{g}) - J(\theta) \geq \gamma\|\widehat{g}\|_2 \left(\|\widehat{g}\|_2 - \varepsilon\right) - \frac{L\gamma^2}{2}\|\widehat{g}\|_2^2.$$

For $\widehat{g} \neq 0$, the right-hand side is a concave quadratic in γ, whose nonnegative maximizer is

$$\gamma = \frac{1}{L}\left(1 - \frac{\varepsilon}{\|\widehat{g}\|_2}\right)_+.$$

Substitution yields

$$J(\theta + \gamma\widehat{g}) - J(\theta) \geq \frac{1}{2L}\left(\|\widehat{g}\|_2 - \varepsilon\right)_+^2.$$

When $\widehat{g} = 0$, both the chosen step and the claimed lower bound are zero. This proves Lemma 1.

C.3 Proof of Lemma 2

If $M_{2,k} = 0$, Equation (21) makes the result immediate. Otherwise, Markov's inequality applied to $\|\widehat{g}_k - g_k\|_2^2$ gives

$$\begin{aligned}\Pr\left(\|\widehat{g}_k - g_k\|_2 > \sqrt{\frac{M_{2,k}}{\delta N \rho_k}}\right) \\ \leq \frac{\mathbb{E}\|\widehat{g}_k - g_k\|_2^2}{M_{2,k}/(\delta N \rho_k)} \leq \delta,\end{aligned}$$

where the last inequality follows from Equation (21). Taking complements proves Equation (7).

C.4 Proof of Theorem 3

By Lemma 2, the event

$$\|\widehat{g}_k - g_k\|_2 \leq \varepsilon_k(\delta) := \sqrt{\frac{M_{2,k}}{\delta N \rho_k}}$$

has probability at least $1 - \delta$. On this event, Lemma 1 applies with $\theta = \theta_k$, $\widehat{g} = \widehat{g}_k$, $\varepsilon = \varepsilon_k(\delta)$, and $L = L_k$. Using the step in Equation (8) gives Equation (9).

C.5 Proof of Corollary 9

Theorem 3 makes the optimization consequence of ESS explicit. Define the positive-certificate event as

$$\mathcal{C}_k := \{\|\widehat{g}_k\|_2 > \varepsilon_k(\delta)\}. \tag{22}$$

Decreasing ESS enlarges the error radius and therefore makes this condition harder to satisfy. The same argument also controls the probability of obtaining an improving update.

Corollary 9 (Positive certificate and actual improvement)**.** *Under the update of Theorem 3,*

$$\Pr(\mathcal{C}_k) - \delta \leq \Pr\{J(\theta_k + \gamma_k\widehat{g}_k) > J(\theta_k)\} \leq \Pr(\mathcal{C}_k). \tag{23}$$

The upper bound follows because $\gamma_k = 0$ whenever $\mathcal{C}_k$ does not occur. On the reliability event of Lemma 2, every batch in $\mathcal{C}_k$ satisfies the strictly positive lower bound in Theorem 3; the complement of this reliability event has probability at most δ, giving the lower bound.

Let

$$\mathcal{R}_k := \{\|\widehat{g}_k - g_k\|_2 \le \varepsilon_k(\delta)\}$$

be the reliability event and recall

$$\mathcal{C}_k := \{\|\widehat{g}_k\|_2 > \varepsilon_k(\delta)\}\,.$$

Lemma 2 gives $\Pr(\mathcal{R}_k) \ge 1 - \delta$.

On $\mathcal{C}_k \cap \mathcal{R}_k$, the lower bound in Theorem 3 is strictly positive. Hence

$$\begin{aligned}\Pr(J(\theta_k + \gamma_k \widehat{g}_k) > J(\theta_k)) &\ge \Pr(\mathcal{C}_k \cap \mathcal{R}_k)\\ &\ge \Pr(\mathcal{C}_k) - \Pr(\mathcal{R}_k^c)\\ &\ge \Pr(\mathcal{C}_k) - \delta.\end{aligned}$$

Conversely, if $\mathcal{C}_k$ does not occur, then $\gamma_k = 0$ by Equation (8); strict improvement is therefore impossible. Thus

$$\Pr(J(\theta_k + \gamma_k \widehat{g}_k) > J(\theta_k)) \le \Pr(\mathcal{C}_k),$$

which proves Corollary 9.

C.5.1 Proof of Lemma 4

Apply Markov's inequality to the nonnegative random variable $\|\widehat{g}_k(u) - g_k\|_2^2$:

$$\Pr\left(\|\widehat{g}_k(u) - g_k\|_2 > \sqrt{\frac{m_k(u)}{\delta}}\right) \le \frac{\mathbb{E}\|\widehat{g}_k(u) - g_k\|_2^2}{m_k(u)/\delta} = \delta.$$

Taking complements proves Equation (10).

C.5.2 Proof of Theorem 5

For any realization of $\widehat{g}_k(u)$, L_k-smoothness gives

$$J(\theta_k + \eta_k \widehat{g}_k(u)) - J(\theta_k) \ge \eta_k g_k^\top \widehat{g}_k(u) - \frac{L_k \eta_k^2}{2}\|\widehat{g}_k(u)\|_2^2.$$

Taking expectations yields

$$\mathbb{E}[J(\theta_k + \eta_k \widehat{g}_k(u)) - J(\theta_k)] \ge \eta_k g_k^\top \mathbb{E}[\widehat{g}_k(u)] - \frac{L_k \eta_k^2}{2}\mathbb{E}\|\widehat{g}_k(u)\|_2^2.$$

It remains to express the right-hand side through the estimator MSE. Expanding $m_k(u)$ gives

$$m_k(u) = \mathbb{E}\|\widehat{g}_k(u)\|_2^2 - 2g_k^\top \mathbb{E}[\widehat{g}_k(u)] + \|g_k\|_2^2,$$

and hence

$$2g_k^\top \mathbb{E}[\widehat{g}_k(u)] = \|g_k\|_2^2 + \mathbb{E}\|\widehat{g}_k(u)\|_2^2 - m_k(u).$$

Substituting this identity into the smoothness bound gives

$$B_k(u;\eta_k) = \frac{\eta_k}{2}\|g_k\|_2^2 - \frac{\eta_k}{2}m_k(u) + \frac{\eta_k}{2}(1 - L_k\eta_k)\mathbb{E}\|\widehat{g}_k(u)\|_2^2,$$

which is Equation (12) and proves the theorem.

C.5.3 Proof of Corollary 6

Apply Equation (12) to u and v. The common term $\eta_k \|g_k\|_2^2/2$ cancels, giving

$$\begin{aligned} B_k(u;\eta_k) - B_k(v;\eta_k) = &\frac{\eta_k}{2}\{m_k(v) - m_k(u)\} \\ &- \frac{\eta_k}{2}(1 - L_k\eta_k)\left\{\mathbb{E}\|\widehat{g}_k(v)\|_2^2 - \mathbb{E}\|\widehat{g}_k(u)\|_2^2\right\}. \end{aligned}$$

Since $\eta_k > 0$, this difference is positive exactly under Equation (13).

C.5.4 Proof of Proposition 7

Set $v = u_{\mathrm{IS}}$ in Corollary 6. From Equation (21),

$$m_k(u_{\mathrm{IS}}) = \frac{1}{N}\left(\frac{M_{2,k}}{\rho_k} - \|g_k\|_2^2\right).$$

Since IS is unbiased,

$$m_k(u_{\mathrm{IS}}) = \mathbb{E}\|\widehat{g}_k(u_{\mathrm{IS}})\|_2^2 - \|g_k\|_2^2,$$

so

$$\mathbb{E}\|\widehat{g}_k(u_{\mathrm{IS}})\|_2^2 = \frac{M_{2,k}}{N\rho_k} + \left(1 - \frac{1}{N}\right)\|g_k\|_2^2.$$

Substituting these two identities into Equation (13) gives exactly Equation (14).

C.5.5 Proof of Proposition 8

For notational brevity, write

$$\widehat{g}_c := \widehat{g}_k(u_{\mathrm{T}}^{(c)}), \qquad \widehat{g}_\infty := \widehat{g}_k(u_{\mathrm{T}}^{(\infty)}).$$

Their difference is

$$\widehat{g}_\infty - \widehat{g}_c = \frac{1}{N}\sum_{i=1}^{N} D_i, \qquad D_i := A_i \sum_{t=1}^{T}(r_{i,t} - c)_+ \ell_{i,t}.$$

Let $H_c < \infty$ uniformly bound the conditional second moments in the statement of the proposition. Using $\|\sum_{t=1}^{T} x_t\|_2^2 \le T\sum_{t=1}^{T}\|x_t\|_2^2$,

$$\begin{aligned} \mathbb{E}\|D_i\|_2^2 &\le T\sum_{t=1}^{T}\mathbb{E}\left[\|A_i(r_{i,t} - c)_+\ell_{i,t}\|_2^2\right] \\ &\le TH_c^2\sum_{t=1}^{T}\Pr(r_{i,t} > c) \\ &= T^2 H_c^2 p_{c,k}. \end{aligned}$$

Jensen's inequality then gives

$$\mathbb{E}\|\widehat{g}_\infty - \widehat{g}_c\|_2^2 \le T^2 H_c^2 p_{c,k} = O(p_{c,k}).$$

To transfer this estimator bound to the policy-improvement certificate, use the equivalent smoothness representation derived in the proof of Theorem 5:

$$B_k(u;\eta_k) = \eta_k g_k^\top \mathbb{E}[\widehat{g}_k(u)] - \frac{L_k\eta_k^2}{2}\mathbb{E}\|\widehat{g}_k(u)\|_2^2.$$

Let $d_c := \widehat{g}_\infty - \widehat{g}_c$. Then

$$B_k(u_{\mathrm{T}}; \eta_k) - B_k(u_{\mathrm{T}}^{(\infty)}; \eta_k) = -\eta_k g_k^\top \mathbb{E}[d_c] \\ + L_k \eta_k^2 \mathbb{E}[\widehat{g}_\infty^\top d_c] - \frac{L_k \eta_k^2}{2} \mathbb{E}\|d_c\|_2^2.$$

By Cauchy–Schwarz,

$$\left| B_k(u_{\mathrm{T}}; \eta_k) - B_k(u_{\mathrm{T}}^{(\infty)}; \eta_k) \right| \le \eta_k \|g_k\|_2 \sqrt{\mathbb{E}\|d_c\|_2^2} \\ + L_k \eta_k^2 \sqrt{\mathbb{E}\|\widehat{g}_\infty\|_2^2 \mathbb{E}\|d_c\|_2^2} \\ + \frac{L_k \eta_k^2}{2} \mathbb{E}\|d_c\|_2^2.$$

The first estimator bound shows that $\mathbb{E}\|d_c\|_2^2 = O(p_{c,k})$. With the remaining moments fixed, the first two terms are therefore $O(\sqrt{p_{c,k}})$ and the last is $O(p_{c,k})$. Hence

$$\left| B_k(u_{\mathrm{T}}; \eta_k) - B_k(u_{\mathrm{T}}^{(\infty)}; \eta_k) \right| = O(\sqrt{p_{c,k}}),$$

as claimed.

C.6 Additional specializations

Intact prompt groups. Suppose M prompts are independent and each prompt has G conditionally independent responses. Keep each prompt group intact and define

$$\Xi_{k,i} := \frac{1}{G} \sum_{j=1}^{G} W_{k,ij} f_{k,ij}.$$

A leave-one-out baseline is admissible because it excludes the focal response; conditional on the prompt and the remaining responses, the focal score still has mean zero. Hence $\mathbb{E}[\Xi_{k,i}] = g_k$.

Jensen's inequality gives

$$\|\Xi_{k,i}\|_2^2 \le \frac{1}{G} \sum_{j=1}^{G} W_{k,ij}^2 \|f_{k,ij}\|_2^2.$$

Repeating the sample-mean argument across the M independent prompt groups therefore yields, under bounded contributions,

$$\mathbb{E} \left\| \frac{1}{M} \sum_{i=1}^{M} \Xi_{k,i} - g_k \right\|_2^2 \le \frac{\|f_k\|_\infty^2}{M \rho_k}.$$

Applying Markov's inequality gives the corresponding version of Lemma 2 with the number of independent prompts M in place of the number of responses N.

Bounded-contribution specialization. If $\|f_k\|_\infty < \infty$, then

$$M_{2,k} = \rho_k \mathbb{E}_Q[W_k^2 \|f_k\|_2^2] \le \|f_k\|_\infty^2.$$

Lemma 2 therefore reduces to the familiar radius

$$\frac{\|f_k\|_\infty}{\sqrt{\delta N \rho_k}}.$$